Astro 797 - Spring 2026

Homework #1 - Due Feb 5th by 10:59am
50 points

Instructions*:* For all answers, show all the steps of your work and explain your reasoning in complete sentences. Clearly mark your final answer to each question. Any numerical substitution into a formula should be clearly indicated. It is best to carry a derivation through symbolically as far as possible and then substitute numbers in at the end.

All work must be handed in as a PDF generated with LaTeX. All coding must be handed in as a Jupyter notebook. I can provide LaTeX templates if you so desire.

You will upload the PDF to the assignment link in Canvas. Any code you write must be uploaded as part of a Jupyter Notebook or Google Colab Notebook. You can upload the notebook or a link to the notebook directly, or make a GitHub repository for this class and assignment and submit the repository address.

While somewhat more time consuming, learning how to document your work this way is good professional practice and will force you (willingly or not) to be more thoughtful about what you write down and how you explain your steps.

Reflection Statement: To obtain any credit you will require a reflection statement at the end of the assignment. This should be 1-2 paragraphs. This reflection statement should include:

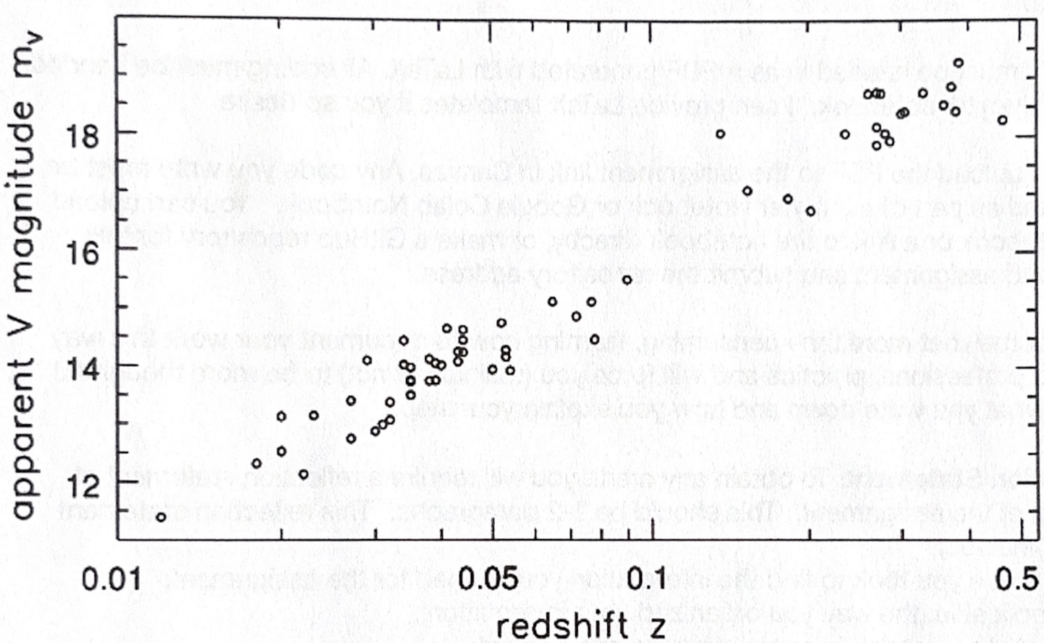
1. the steps you took to find the information you needed for the assignment;
2. if applicable, the way you organized your information;
3. the steps you took in completing the assignment;
4. If you used generative AI, explain which tool you used, how you used it, and to what extent. Also explain what you learned from the process. Were AI tools helpful? If so, what approaches do you want to use next time and what approaches should you avoid?
5. Did the use of generative AI make the assignment feel less like your own work? Or were you able to edit and use other strategies to shape the work into your own?

Advice*:* Start early so that you can ask me if you are stuck on a problem.

1.

a. If a galaxy has absolute magnitude M, derive an expression for the apparent magnitude m in terms of the redshift $z=V_r/c$, the absolute magnitude M, and a constant C, which is the same for all objects.

b. Draw an approximate straight line through the points in the figure below, which shows the apparent magnitude for the brightest galaxy in a galaxy cluster and calculate its slope. Compare its slope to what you would expect if the brightest galaxy in a rich cluster always has the same luminosity.



2. The ALMA radio telescope can observe over a simultaneous frequency range of 8GHz. Assume that the frequency of the ALMA instrument (called a "correlator") can be centered at any observed frequency, e.g. that of a galaxy at rest in a cluster.

The CO(2-1) spectral line has a rest-frame frequency of 230.538 GHz. You would like to observe this line coming from galaxies in a cluster at z=1.600. How fast can a galaxy be moving along the line of sight with respect to the center of the cluster such that you will still be able to detect the CO line? Express your answer in km/s.

3. Figure 6 from van der Wel et al. 2014, ApJ, 788, 28 shows in the left panel the **radius** of a galaxy with a stellar mass of $M_\star = 5 \times 10^{10} M_\odot$ as a function of redshift. The red and blue points in that plot correspond to galaxies that are not forming stars (passive) or are forming stars (star forming) respectively. Using the parametric fit to these data as a function of $(1+z)$, compute the angular **diameter** of galaxies with $M_\star = 5 \times 10^{10} M_\odot$ over the full range $z = 0 - 3$. Assume a $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, and $h = 0.7$ cosmology.

   I want you to write a program to solve this problem by numerically integrating *E(z)*. Don't use the astropy.cosmology package nor should you use Ned Wright's cosmology calculator. You can, use any numerical method to integrate E(z) though the trapezoid method will suffice.

   You must include your code with this assignment, which you can upload as a jupyter notebook, link to a Google Colab Notebook, or a standalone python code in a GitHub repository. This code must be well commented. You **cannot** use generative AI tools to help you with the coding part of this assignment.

   Note: The radius here is that which encloses 50% of the light, as is standard practice. We will adopt this as the radius of the galaxy for the purpose of this problem.

4. Compute the comoving volume in the three intervals $0.05 < z \leq 1$, $1 < z \leq 2$, and $2 < z \leq 3$ that is contained within a field whose solid angle is 0.25 deg². Assume $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, and $h = 0.7$. I want you to write a program to solve this problem by numerically integrating *E(z)*. Don't use the astropy.cosmology package nor should you use Ned Wright's cosmology calculator. You may want to refer to "Distance Measures in Cosmology" by Hogg for an alternate formulation of the Volume element.

   As part of your answer, briefly discuss the the difference in the enclosed volume in the three intervals, all which have nearly identical Δz.

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